

# CASE 8

## MHC CLASS II DEFICIENCY

### An inherited failure of gene regulation.

The class II molecules of the major histocompatibility complex (MHC) are involved in presenting antigens to CD4<sup>+</sup> T cells. The peptide antigens that they present are derived from extracellular pathogens and proteins taken up into intracellular vesicles, or from pathogens such as *Mycobacterium* that persist intracellularly inside vesicles. MHC class II molecules are expressed constitutively on antigen-presenting cells, including dendritic cells, monocyte/macrophages, and B lymphocytes. In humans, MHC class II molecules, together with MHC class I molecules (see Case 12), are known as the HLA antigens. They are also expressed on the epithelial cells of the thymus and their expression can be induced on other cells, principally by the cytokine interferon- $\gamma$ . T cells also express MHC class II molecules when they are activated.

MHC class II molecules are heterodimers consisting of an  $\alpha$  chain and a  $\beta$  chain (Fig. 8.1). The genes encoding both chains are located in the Human Leukocyte Antigen (HLA) locus on the short arm of chromosome 6 in humans (Fig. 8.2). The principal MHC class II molecules are designated DP, DQ, and DR, and, like the MHC class I molecules, they are highly polymorphic. Peptides bound to MHC class II molecules can be recognized only by the T-cell receptors of CD4<sup>+</sup> T cells and not by those of CD8<sup>+</sup> T cells (Fig. 8.3). MHC class II molecules expressed in the thymus also have a vital role in the intrathymic maturation of CD4<sup>+</sup> T cells.

Expression of the genes encoding the  $\alpha$  and  $\beta$  chains of MHC class II molecules must be strictly coordinated and it is under complex regulatory control by a series of transcription factors. The existence of these transcription factors and a means of identifying them were first suggested by the study of patients with MHC class II deficiency.

#### TOPICS BEARING ON THIS CASE:

Role of MHC class II molecules in antigen presentation to CD4 T cells

Role of co-receptor molecule CD4 in antigen recognition by T cells

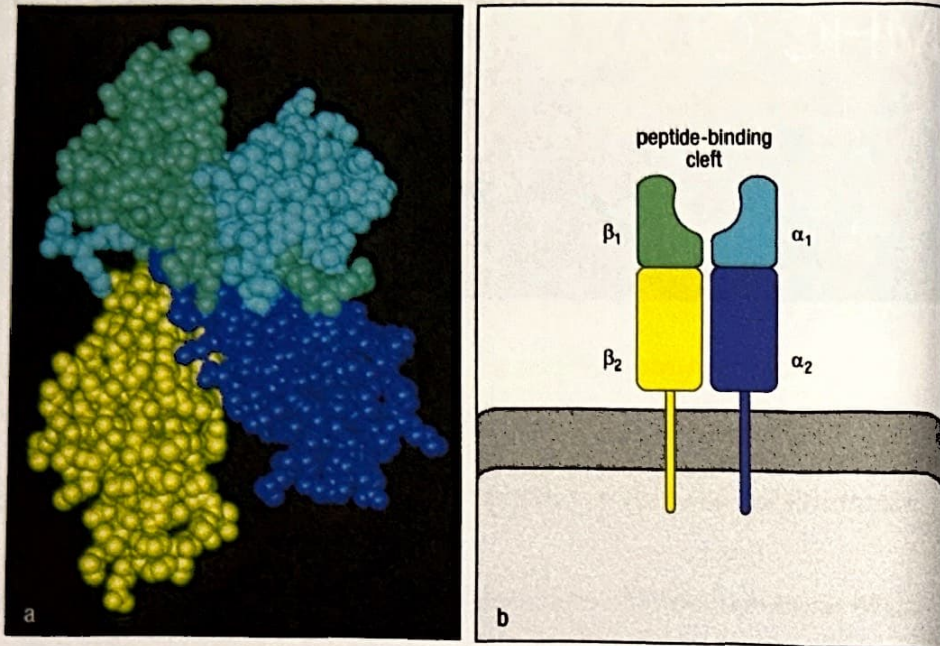
Intrathymic maturation of CD4 T cells

Mixed lymphocyte reaction

Lymphocyte stimulation by polyclonal mitogens

FACS analysis

**Fig. 8.1 Structure of an MHC class II molecule.** Panel a shows a computer graphic representation of the MHC class II molecule HLA-DR1. Panel b is a schematic representation of the molecule. It is composed of two transmembrane glycoprotein chains,  $\alpha$  and  $\beta$ , each folded into two protein domains. The antigenic peptide binds in a cleft between the two chains. Photograph courtesy of C. Thorpe.



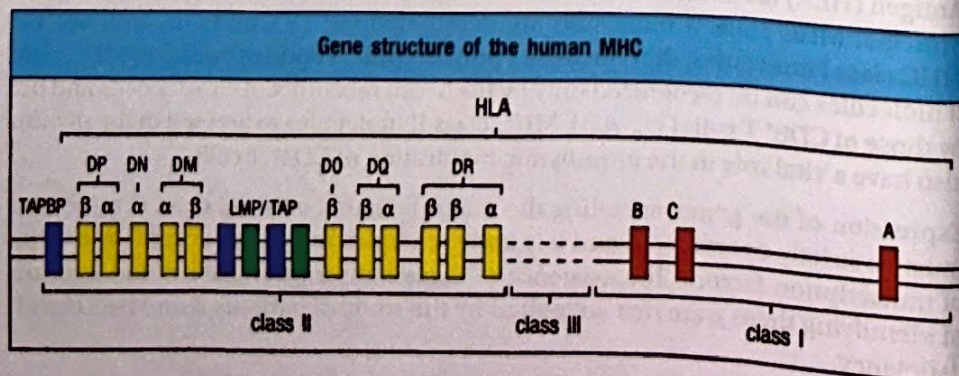
### The case of Helen Burns: a 6-month-old child with a mild form of combined immunodeficiency.

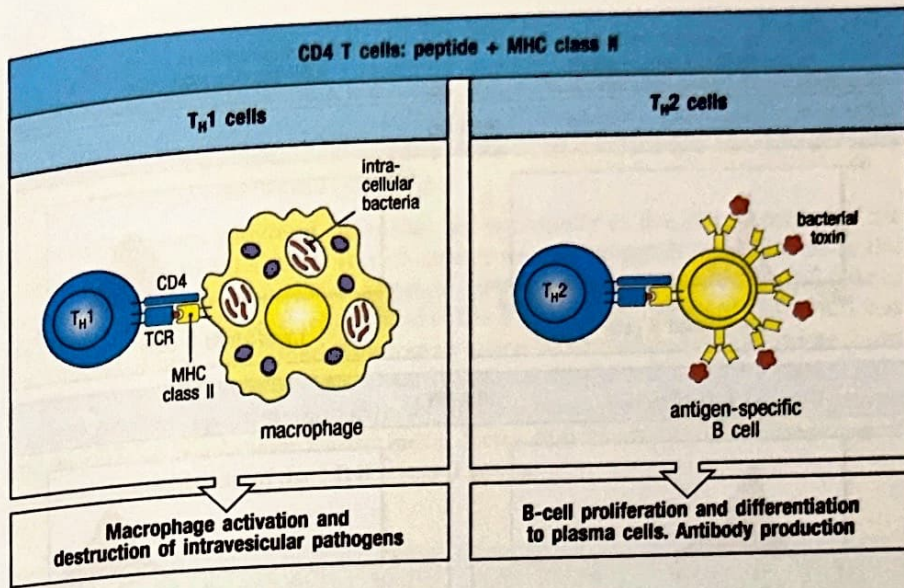
Helen Burns was the second child born to her parents. She thrived until 6 months of age when she developed pneumonia in both lungs, accompanied by a severe cough and fever. Blood and sputum cultures for bacteria were negative, but a tracheal aspirate revealed the presence of abundant *Pneumocystis jirovecii*. She was treated successfully with intravenous administration of high doses of trimethoprim-sulfamethoxazole, which is very effective against *Pneumocystis*, and seemed to recover fully.

As her pneumonia was caused by the opportunistic pathogen *P. jirovecii*, Helen was suspected to have severe combined immunodeficiency. A blood sample was taken, and her peripheral blood mononuclear cells were stimulated with phytohemagglutinin (PHA) to test for T-cell function by  $^3\text{H}$ -thymidine incorporation into DNA. A normal T-cell proliferative response was obtained, with her T cells incorporating  $114,050 \text{ counts min}^{-1}$  of  $^3\text{H}$ -thymidine (normal control  $75,000 \text{ counts min}^{-1}$ ). Helen had received routine immunizations at 2 months of age. However, in further tests her T cells failed to respond to tetanus toxoid *in vitro*, although they responded normally in the  $^3\text{H}$ -thymidine incorporation assay when stimulated in a mixed leukocyte reaction (see Fig. 6.3).

6-month-old girl with pneumonia. SCID? Do lymphocyte function tests.

**Fig. 8.2 The organization of the major histocompatibility complex (MHC) on chromosome 6 in humans.** There are separate regions of class I and class II genes. The class I genes encode for the HLA-A, HLA-B, and HLA-C proteins. The class II genes are shown in yellow.





**Fig. 8.3 Effector CD4<sup>+</sup> cells recognize antigens bound to MHC class II molecules.** CD4<sup>+</sup> T cells carry the co-receptor molecule CD4, which binds to MHC class II molecules on the antigen-presenting cell and helps to stabilize the binding of T-cell receptor to antigen. Effector CD4 T cells fall into several different subclasses: TH1 and TH2 are shown here. TH1 cells are involved mainly in responding to antigens presented by macrophages, which activates the macrophages causing destruction of the bacteria within the macrophage vesicles. TH2 cells respond to antigen presented by B cells, stimulating the differentiation of B cells to plasma cells and the production of antibodies.

When it was found that Helen's T cells could not respond to a specific antigenic stimulus, her serum immunoglobulins were measured and found to be very low. IgG levels were 96 mg dl<sup>-1</sup> (normal 600–1400 mg dl<sup>-1</sup>), IgA was 6 mg dl<sup>-1</sup> (normal 60–380 mg dl<sup>-1</sup>), and IgM 30 mg dl<sup>-1</sup> (normal 40–345 mg dl<sup>-1</sup>).

Helen's white blood cell count was elevated at 20,000 cells  $\mu$ l<sup>-1</sup> (normal range 4000–7000  $\mu$ l<sup>-1</sup>). Of these, 82% were neutrophils, 10% lymphocytes, 6% monocytes, and 2% eosinophils. The calculated number of 2000 lymphocytes  $\mu$ l<sup>-1</sup> was low for her age (normal >2500  $\mu$ l<sup>-1</sup>). Of her lymphocytes, 27% were B cells as determined by an antibody against CD20 (normal 10–12%), and 47% reacted with antibody to the T-cell marker CD3. In particular, 41% of Helen's lymphocytes were positive for CD8, and 6% were positive for CD4. Thus, at 810 cells  $\mu$ l<sup>-1</sup> her number of CD8<sup>+</sup> T cells was within the normal range, but the number of CD4<sup>+</sup> T cells (120  $\mu$ l<sup>-1</sup>) was much lower than normal (her CD4<sup>+</sup> T-cell count would be expected to be twice her CD8<sup>+</sup> T-cell count). The presence of substantial numbers of T cells, and thus a normal response to PHA, ruled out a diagnosis of severe combined immunodeficiency (see Case 5).

Helen's pediatrician referred her to the Children's Hospital for consideration for a bone marrow transplant, despite the lack of a diagnosis. When an attempt was made to HLA-type Helen, her parents, and her healthy 4-year-old brother by serology, a DR type could not be obtained from Helen's white blood cells. Her circulating B lymphocytes and her monocytes did not express HLA-DR molecules. Hence, a diagnosis of MHC class II deficiency was established (Fig. 8.4). Ultimately, HLA typing was performed by molecular DNA techniques.

Her brother was found to have the same HLA type as Helen, and therefore was chosen as a bone marrow donor. Helen was given 1 mg kg<sup>-1</sup> of body weight of the cytotoxic drug busulfan every 6 hours for 4 days and then 50 mg kg<sup>-1</sup> cyclophosphamide each day for 4 days to ablate her bone marrow. The brother's bone marrow was administered to Helen by transfusion without any *in vitro* manipulation. The graft was successful and immune function was restored.

## MHC class II deficiency.

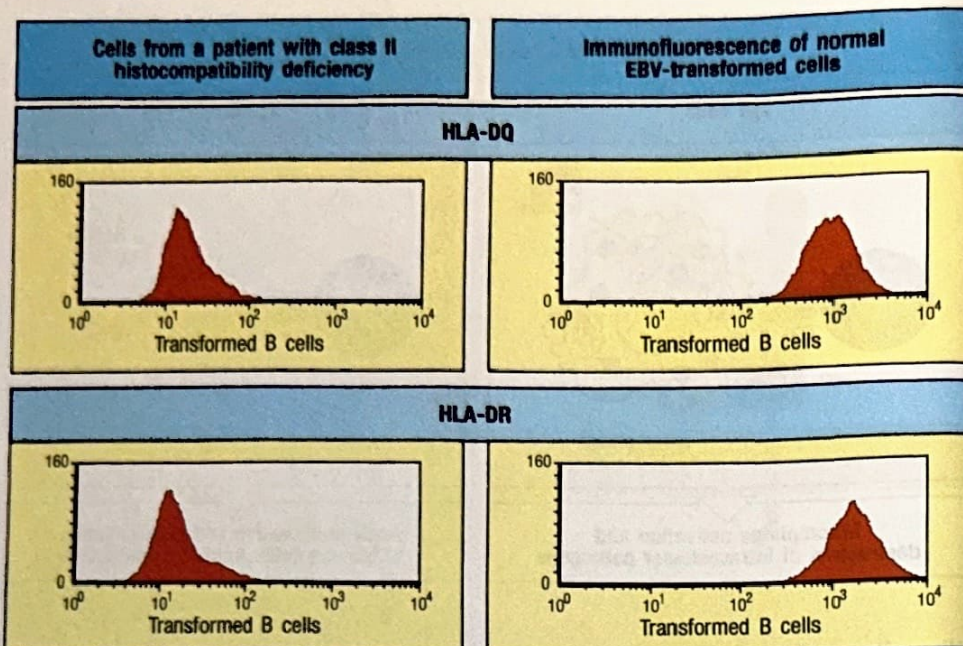
MHC class II deficiency is inherited as an autosomal recessive trait. Health problems show up early in infancy. Affected babies present the physician with combined immunodeficiency as they have increased susceptibility to pyogenic and

*Low Ig levels, deficiency of CD4 T cells.*

*No HLA-DR type available. Do FACS analysis.*

*MHC class II deficiency. Bone marrow transplant advisable, but results often unsatisfactory.*

**Fig. 8.4 Detection of MHC class II molecules by fluorescent antibody.** Helen's transformed B-cell line was examined by using a fluorescent antibody against HLA-DQ and HLA-DR. Helen (left panels) expressed approximately 1% of the amount of MHC class II molecules compared with a transformed B-cell line from a normal control (right panels).



opportunistic infections. However, they differ from infants with severe combined immunodeficiency (SCID; see Case 5) in that they have T cells, which can respond to nonspecific T-cell mitogens such as PHA and to allogeneic stimuli. Unlike in some other types of immunodeficiency, progressive infection with the attenuated live vaccine strain *Bacillus Calmette-Guérin* (BCG) has not been observed in MHC class II-deficient patients after BCG vaccination against tuberculosis (most cases of MHC class II deficiency have been observed in North African migrants in Europe, where BCG vaccination is routine). This is because mycobacterial antigens derived from BCG can be presented on MHC class I molecules and infected cells can be destroyed by cytotoxic T cells. In contrast, and for reasons that are unclear so far, patients with MHC class II deficiency are highly prone to severe viral infections.

Patients with MHC class II deficiency are deficient in  $CD4^+$  T cells, in contrast with MHC class I deficiency, in which  $CD8^+$  T-cell numbers are very low and the levels of  $CD4^+$  T cells are normal (see Case 12). Typically, patients with MHC class II deficiency also have moderate to severe hypogammaglobulinemia.

Hematopoietic stem cell transplantation (HSCT) is the treatment of choice for patients with MHC class II deficiency. Helen Burns was cured after a bone marrow transplant from her HLA-identical brother. However, the results of HSCT in patients with MHC class II deficiency, even when transplanted from HLA-identical donors, are often not satisfactory, and the number of circulating  $CD4^+$  T cells frequently remains low. This is likely because positive selection of donor-derived  $CD4^+$  thymocytes is compromised, owing to a lack of MHC class II molecules on the surface of the patient's thymic epithelial cells.

Genetic linkage analysis in large extended families with MHC class II deficiency has shown that this condition is not linked to the MHC locus on the short arm of chromosome 6 and that the genes encoding the MHC class II molecules at this locus are normal. Interferon- $\gamma$  induces the expression of MHC class II molecules on antigen-presenting cells from normal people but fails to induce their expression on the antigen-presenting cells of patients with MHC class II deficiency. This suggested that the defect might lie in the regulation of expression of the MHC class II genes.

The search for the cause of the defect was complicated further by the discovery that MHC class II deficiency in different patients seems to have different causes. Epstein-Barr virus (EBV)-transformed B cells isolated from class II-deficient patients do not express MHC class II molecules. However, when B cells from two

different patients are fused, MHC class II expression is often observed. The fusion of the two cell lines has corrected the defect. This means that one cell must be able to replace whatever is lacking in the other, and thus the two cells must carry different genetic defects causing the MHC class II deficiency. Pairwise fusions were performed on a large number of cell lines from different patients, and four complementation groups were found (Fig. 8.5).

These experiments provided clues that led eventually to the identification of the defect. The lack of MHC class II molecules turns out to result from defects in the transcription factors required to regulate their coordinated expression. All four of these transcription factors, which bind to the 5' regulatory region of the MHC class II genes, have been identified. The transcription factors whose mutations account for MHC class II deficiency are Class II Transactivator (CIITA), RFXANK, RFX5, and RFXAP. The last three bind directly to the promoter of all MHC class II genes, whereas CIITA is a master transcription factor that binds the RFX complex and activates transcription of the MHC class II genes.

## Questions.

- 1 Why did Helen lack CD4 T cells in her blood?
- 2 Why did Helen have a low level of immunoglobulins in her blood?
- 3 In SCID, lymphocytes fail to respond to mitogenic stimuli. Although Helen was first thought to have SCID, this diagnosis was eliminated by her normal response to PHA and an allogeneic stimulus. How do you explain these findings?
- 4 If a skin graft were to be placed on Helen's forearm do you think she would reject the graft?

|   | A | B | C | D |
|---|---|---|---|---|
| A | — | + | + | + |
| B | + | — | + | + |
| C | + | + | — | + |
| D | + | + | + | — |

**Fig. 8.5 Complementation groups of MHC class II deficiency.** B-cell lines isolated from different patients were fused in all pairwise combinations to determine whether they could correct each other's defect. If two cell lines do not correct each other (—), they are in the same complementation group and have the same genetic defect. However, if the defect is corrected (+), the two cell lines belong to two different complementation groups and have two different defects. Four complementation groups, A, B, C, and D, were discovered by this technique.